

SmartVision_AI using Deep Learning & Computer Vision

1. Introduction

SmartVision_AI is a Computer Vision application built using Deep Learning, Transfer Learning, YOLOv8m, and Streamlit. The system performs Image Classification and Object Detection using a curated subset of the COCO dataset.

2. Problem Statement

Traditional image recognition systems require large datasets and extensive computation. This project aims to classify images into object categories and detect multiple objects efficiently through an interactive application.

3. Objectives

- Build an image classification system using Transfer Learning
- Compare multiple CNN models
- Implement YOLOv8m for object detection
- Develop an interactive Streamlit application
- Evaluate performance using standard metrics

4. Dataset Description

The project uses a curated subset of the COCO dataset containing 2,500 images across 26 categories (100 images per class). The data was split into Training (70%), Validation (15%), and Testing (15%).

5. Models Used

Classification Models: ResNet50, VGG16, MobileNetV2, EfficientNetB0

Object Detection: YOLOv8m

6. Model Performance

ResNet50 achieved the best classification performance with an accuracy of **55.25%**. MobileNetV2 offered lightweight and efficient inference, while YOLOv8m successfully performed object detection with bounding boxes and confidence scores.

7. Streamlit Application

The application includes:

- Home page with project overview
- Image Classification page for prediction comparison
- Object Detection page using YOLOv8m
- Model Performance dashboard

8. Challenges Faced

- **Dataset Quality:** COCO images included duplicate, blackened, corrupted, and poor-quality samples, affecting classification accuracy.
- **GPU Limitations:** Deep learning training required significant computation and increased training time.
- **Dependency Compatibility:** TensorFlow/Keras version issues occurred during deployment.
- **Large Model Management:** Git LFS was required to manage large model files.

9. Conclusion

SmartVision_AI successfully integrates image classification and object detection into a unified AI system. The project demonstrates the practical application of Deep Learning and Computer Vision techniques using Transfer Learning and YOLOv8m.

10. Future Improvements

- Real-time webcam detection
- More categories and improved dataset quality
- Cloud deployment optimization
- Faster inference and enhanced accuracy

11. Tech Stack

Python, TensorFlow, PyTorch, YOLOv8, Streamlit, OpenCV, NumPy, Pandas, Matplotlib, Scikit-learn

12. Author

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